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# St. Joseph's College of Arts & Science (Autonomous)
## St. Joseph's College Road, Cuddalore – 607001

**CH102Q – KINETIC THEORY OF GASES AND CHEMICAL KINETICS**

**Time : 3 hrs** **Max Marks :75**

## SECTION – A (20X1=20)

Answer **ALL** Questions

### I. Choose the correct answer

1. The value for gas constant 'R' is ---------- for the unit Latm $K^{-1}mol^{-1}$
   a) 0.0821 b) 82.1 c) 8.314 d) $8.314X10^{7}$

2. V∞n (n,P are constant) is ---------- law
   a) Boyle's b) Charle's c) Molefraction d) Avogadro's

3. -------- is the temperature at which the second virial coefficient vanishes.
   a) Critical temperature b) Eutectic point
   c) Boiling point d) Boyle temperature

4. $(8RT/\pi M)^{½}$ is the formula used to find ---------- velocity.
   a) RMS b) Most probable c) Molar d) Average

5. For a spontaneous reaction, ΔG=------
   a) +ve b) -ve c) 0 d) Any value

6. For the reaction $PCl_5(g) \leftrightarrow PCl_3(g)+Cl_2(g)$; Kp in SI units is given by-------
   a) Pa b) Pa-1 c) atm d) atm-1

7. The number of reactant molecules involved in a reaction is -------
   a) Rate b) Molecularity c) Order d) Inversion

8. For a zero order reaction A→Products, t½ is proportional to ---------
   a) [A] b) 1/[A] c) $[A]^2$ d) $1/[A]^2$

9. In a solution of $CO_2$ in water, $CO_2$is ---------.
   a) Solute b) Solvent c) Colloid d) Solution

10. ------- are intermediate between true solutions and suspension
    a) Colloids b) Precipitate c) Solution d) Solutes

**II. Match the following**

11. 1 cal - z=1 for ideal gas

12. Compressibility factor - 4.184J

13. k1/k2 - $dm^{-3}$ $mols^{-1}$

14. ko - Parallel layers

15. Smectic - Equilibrium concentration

**SECTION – B (10X2=20)**

Answer any **TEN** Questions

16. What is Virial equation of state?

17. What is Dalton's law of effusion?

18. What is Joule Thomson effect?

19. Write a notes on molecular collision.

20. What is Average velocity?

21. Calculate Kp for the reaction $CO(g)+H_2O(g)\leftrightarrow CO_2+H_2(g)$ at 298K.Calculate $\Delta G^o$.

22. What is Le Chatlier's principle?

23. Calculate the units of the rate constants for i)2, ii) ½ order .

24. What is average rate of a reaction?

25. What are the types in expressing the concentration of solutions?

26. Classify liquid crystals with examples.

27. What is electrical double layer?

## SECTION- C (5X8=40)

Answer any **FIVE** Questions

28. a. Derive Perfect gas equation. **(4)**
    b. Explain Boyle temperature. **(2)**
    c. Write Vander Waals equation of state. **(2)**

29. a. Calculate the most probable velocity of $N_2$ at $15^oC$. **(3)**
    b. Derive the relationship between most probable and RMS velocity. **(3)**
    c. What is Collision diameter? **(2)**

30. a. Derive the relationship between Kp&Kc. **(4)**
    b. Explain BET theory of adsorption. **(4)**

31. a. Explain the effect of change of concentration in equilibrium. **(5)**
    b. What are the applications of Adsorption? **(3)**

32. a. Explain the effect of catalyst on the rate of a reaction with graph. **(5)**
    b. What is half-life of a reaction? **(3)**

33. a. For a $1^{st}$ order reaction is on one fifth completed in 40 mins. How long will it take for the reaction to get completed 100%? **(5)**
    b. Explain temperature coefficient. **(3)**

34. a. Calculate the molefraction of HCl in a solution of hydrochloric, containing 36% HCl by Weight. **(4)**
    b. Explain the characteristics of Lyophilic and Lyophobic solutions. **(4)**

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